

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 7

Name:	___Muhammad_Usman___		Roll number:	___23BSCYS006___	
Score:	_____	Date:	_____	Instructor's Signature:	_____

Lab # 7: To Access the Internet Using IPv4 PPPoE on the Firewall

Objective: The objective of this lab is to access the Internet using IPv4 PPPoE on the Firewall

Duration: 3 Hours

Expected Outcomes: By the end of this lab session, students will be able to:

1. Provide Dynamic IP Allocation
2. Enable Secure Internet Access
3. Establish Carrier Connectivity via PPPoE

Rubrics

LAB PERFORMANCE INDICATOR	SUBJECT KNOWLEDGE	DATA ANALYSIS AND INTERPRETATION	ABILITY TO CONDUCT EXPERIMENT
SCORE			

Note: Student performance will be assessed based on this rubric during lab sessions. **Scenario**

This section provides an example for the device, working as a PPPoE client, to obtain an IP address by dialing up to the carrier's server through PPPoE for Internet access.

What is PPPoE?

PPPoE is a **network protocol** that encapsulates **PPP (Point-to-Point Protocol)** packets inside **Ethernet frames**.

It is mainly used by **Internet Service Providers (ISPs)** to deliver broadband Internet services over Ethernet networks (DSL, Fiber, etc.).

In simple terms: **PPPoE combines the authentication, encryption, and session features of PPP with the physical transport of Ethernet.**

Why PPPoE is Used?

- ISPs need a way to **authenticate users** before giving them Internet access.
- PPP already provides **authentication (PAP/CHAP)** and session management.

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 7

- Ethernet is widely used in LANs but does not provide authentication.
 - PPPoE = PPP + Ethernet → Authentication + IP Address Assignment + Easy Transport
-

PPPoE Workflow (Step by Step)

The PPPoE process has **two main phases**:

1. **Discovery Phase**
 2. **Session Phase**
-

1. Discovery Phase (Finding and Establishing Session)

During this phase, the client (PC or router) discovers the PPPoE server (usually at the ISP). It happens in **4 steps (PADI → PADO → PADR → PADS)**:

1. **PADI (PPPoE Active Discovery Initiation)** ○ Client broadcasts a request asking: *“Is there any PPPoE server available?”*
2. **PADO (PPPoE Active Discovery Offer)** ○ ISP’s PPPoE server replies: *“Yes, I am available to provide service.”*
3. **PADR (PPPoE Active Discovery Request)** ○ Client sends a request to one of the servers: *“I choose you. Please start session.”*
4. **PADS (PPPoE Active Discovery Session-confirmation)** ○ Server confirms: *“Session established.”* and assigns a **Session ID**.

At this point, the **discovery is done**, and the session starts.

2. Session Phase (Authentication + Data Transfer)

- After session establishment, **PPP comes into play**:
 - Authentication (using **PAP** or **CHAP**)
 - Assignment of IP Address (from ISP’s DHCP or RADIUS)
- Once authenticated, **user traffic is carried inside PPPoE packets** over Ethernet.
- The session continues until terminated by either the client or server.

Key Features of PPPoE

- **Authentication:** Supports PAP/CHAP authentication.
- **Per-user accounting:** ISPs can track usage (time, data, bandwidth).
- **Session control:** Each user has a unique session ID.
- **Compatibility:** Works with Ethernet, DSL, and Fiber connections.

Networking Requirements

As shown in [Figure 1](#), FW works as an egress gateway, providing an Internet egress for PCs on the LAN. The company network is planned as follows:

- All PCs on the LAN are deployed on network segment 10.1.1.0/24, and they dynamically obtain IP addresses through DHCP.
- The device connects to all PCs of the company over the downstream link.
- The device applies for Internet service from the carrier over the upstream link. The Internet access service is provided using the PPPoE protocol.

According to the previous requirements, specify the FW as a PPPoE client. After the client obtains IP and DNS addresses from the carrier's server, the Intranet users can access the Internet.

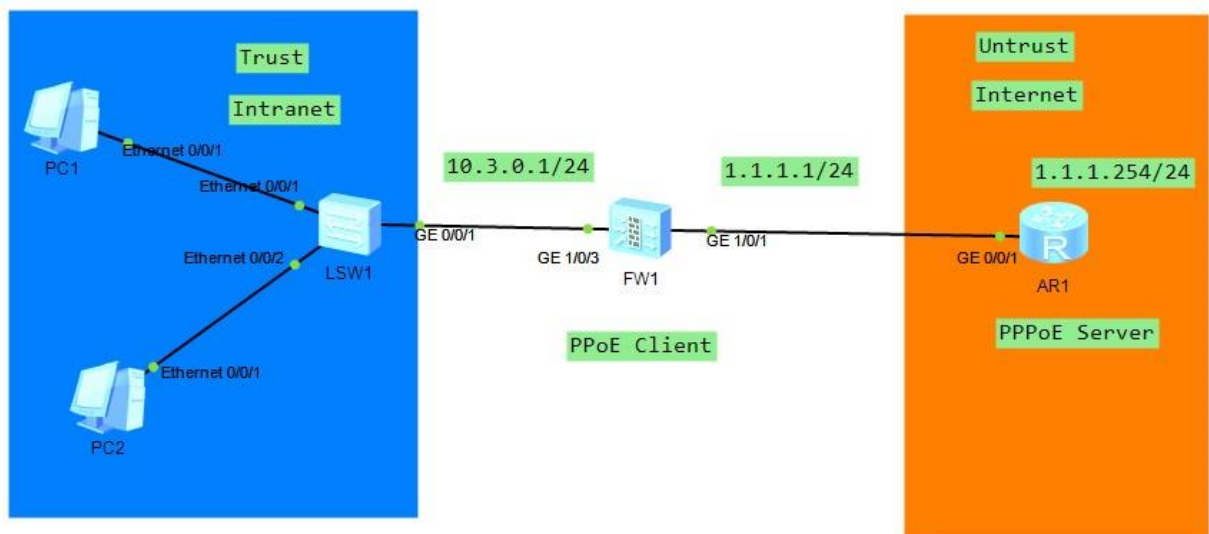


Figure 1 Networking diagram of accessing the Internet through PPPoE

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 7

Table 1: Parameters provided by an ISP

Item	Description
Interface number: GigabitEthernet 1/0/1 Security zone: untrust	The device obtains IP and DNS addresses from the PPPoE server (deployed by the carrier) through dial-up. - Dial-up user name: user - Dial-up password: password
Interface number: GigabitEthernet 1/0/3 IP address: 10.3.0.1/24 Security zone: Trust	DHCP is used to dynamically assign IP addresses to PCs on the LAN.

Configuration Roadmap

1. Configure the downstream link.
Enable DHCP server on the **GigabitEthernet 1/0/3** interface so that it dynamically assigns IP addresses to PCs, and specify the **GigabitEthernet 1/0/3** interface's IP address as the gateway and DNS server addresses for the PCs.
During Internet access, a PC usually requires domain name resolution. This is why a DNS server shall be specified. In this example, FW works as a DNS relay.
2. Configure the upstream link and use **PPPoE** to obtain IP and DNS addresses.
3. Add the interfaces into **security zones** and configure **security policies**. Add the interface connected to the LAN to a **high-priority security zone (Trust zone)**, and the upstream interface connected to the Internet to a **low-priority security zone (Untrust zone)**.
4. The IP addresses used on LANs are private IP addresses, which shall be converted by NAT to public IP addresses for Internet access, if needed. In this

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 7

example, the upstream interface obtains its IP address by **dial-up**. This IP address may vary for each dial-up. Therefore, **easy IP** is recommended.

Procedure

1. Configure the IP address of the interface GigabitEthernet 1/0/3.

```
<FW> system-view
[FW] interface GigabitEthernet 1/0/3
[FW-GigabitEthernet1/0/3] ip address 10.3.0.1 255.255.255.0
[FW-GigabitEthernet1/0/3] quit
```

2. Assign the interfaces to the security zones.

```
[FW] firewall zone untrust
[FW-zone-untrust] add interface GigabitEthernet 1/0/1
[FW-zone-untrust] quit
[FW] firewall zone trust
[FW-zone-trust] add interface GigabitEthernet 1/0/3
[FW-zone-trust] quit
```

3. Configure the device as a DHCP server to assign IP addresses to PCs on the LAN.

Enable the DHCP function.

```
[FW] dhcp enable
```

Create an interface address pool on the interface and specify the DNS server for the Intranet PCs.

```
[FW] interface GigabitEthernet 1/0/3
[FW-GigabitEthernet1/0/3] dhcp select interface
[FW-GigabitEthernet1/0/3] dhcp server ip-range 10.3.0.1 10.3.0.254
[FW-GigabitEthernet1/0/3] dhcp server dns-list 9.9.9.9
[FW-GigabitEthernet1/0/3] dhcp server gateway-list 10.3.0.1
[FW-GigabitEthernet1/0/3] quit
```

4. Configure interface GigabitEthernet 1/0/1 so that it obtains IP and DNS addresses using PPPoE.

```
[FW] dialer-rule 1 ip permit
[FW] interface Dialer 1
[FW-Dialer1] link-protocol ppp
[FW-Dialer1] dialer user user
```

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 7

```
[FW-Dialer1] ip address ppp-negotiate
[FW-Dialer1] ppp ipcp dns admit-any
[FW-Dialer1] dialer-group 1
[FW-Dialer1] dialer bundle 1
[FW-Dialer1] ppp pap local-user user password cipher password
[FW-Dialer1] quit
[FW] firewall zone untrust
[FW-zone-untrust] add interface Dialer 1
[FW-zone-untrust] quit
[FW] interface GigabitEthernet 1/0/1
[FW-GigabitEthernet1/0/1] pppoe-client dial-bundle-number 1 ipv4
[FW-GigabitEthernet1/0/1] quit
```

5. Configure a security policy to allow Intranet PCs to access the Internet.

```
[FW] security-policy
[FW-security-policy] rule name policy_sec_1
[FW-security-policy-policy_sec_1] source-address 10.3.0.0 mask 255.255.255.0
[FW-security-policy-policy_sec_1] source-zone trust
[FW-security-policy-policy_sec_1] destination-zone untrust
[FW-security-policy-policy_sec_1] action permit
[FW-security-policy-policy_sec_1] quit
[FW-security-policy] quit
```

6. Configure a NAT policy to allow Intranet users to access the Internet.

```
[FW] nat-policy
[FW-policy-nat] rule name policy_nat_1
[FW-policy-nat-rule-policy_nat_1] source-address 10.3.0.0 mask 255.255.255.0
[FW-policy-nat-rule-policy_nat_1] source-zone trust
[FW-policy-nat-rule-policy_nat_1] egress-interface GigabitEthernet 1/0/1
[FW-policy-nat-rule-policy_nat_1] action source-nat easy-ip
[FW-policy-nat-rule-policy_nat_1] quit
[FW-policy-nat] quit
```

7. Configure a default route to ensure that the LAN users are routable to the Internet. The next hop shall be the gateway address assigned by the carrier to the enterprise.

```
[FW] ip route-static 0.0.0.0 0.0.0.0 Dialer 1
```

Task:

1. Ping from PC1 to the internet on the given ip address 1.1.1.254
2. Ping from PC2 to the internet on the given ip address 1.1.1.254

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 7

3. Attach Screenshots

Note: Attach all the screenshots of the configuration. The Firewall name should be your roll number like FW1-22BSCYS001. Upload the .topo and .cfg files on MS Teams Assignment.

1. PPPoE Client and Server

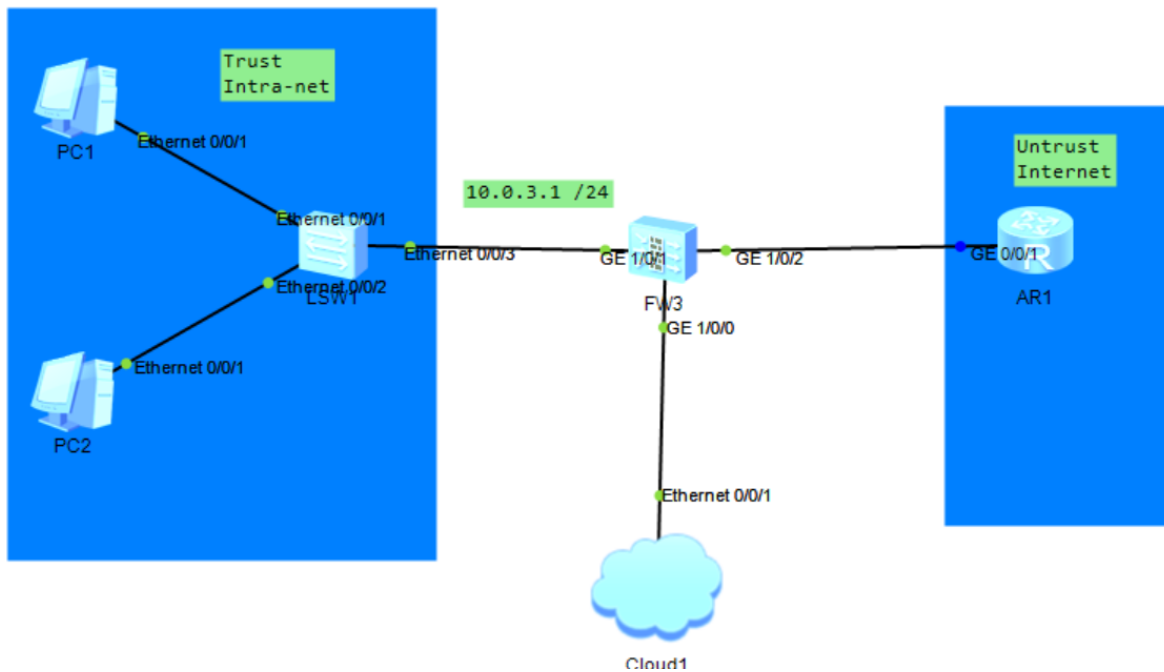
- PPPoE (Point-to-Point Protocol over Ethernet) is used to connect a user to the internet through a network.
- PPPoE Client: The device (like a computer or router) that requests an internet connection using a username and password.
- PPPoE Server: The device (usually at the ISP side) that authenticates the client and provides internet access.

2. PAP and CHAP

Both are authentication methods used to verify a user's identity.

- PAP (Password Authentication Protocol): Sends the password in simple/plain form (less secure).
- CHAP (Challenge Handshake Authentication Protocol): Uses encrypted verification and checks the user repeatedly (more secure).

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 7



DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 7

PC1

Basic Config	Command	MCPacket	UdpPacket	Console
<pre> From 1.1.1.254: bytes=32 seq=3 ttl=254 time=31 ms From 1.1.1.254: bytes=32 seq=4 ttl=254 time=47 ms From 1.1.1.254: bytes=32 seq=5 ttl=254 time=31 ms --- 1.1.1.254 ping statistics --- 5 packet(s) transmitted 5 packet(s) received 0.00% packet loss round-trip min/avg/max = 31/53/109 ms PC>ping 1.1.1.254 Ping 1.1.1.254: 32 data bytes, Press Ctrl_C to break From 1.1.1.254: bytes=32 seq=1 ttl=254 time=31 ms From 1.1.1.254: bytes=32 seq=2 ttl=254 time=32 ms From 1.1.1.254: bytes=32 seq=3 ttl=254 time=46 ms From 1.1.1.254: bytes=32 seq=4 ttl=254 time=32 ms From 1.1.1.254: bytes=32 seq=5 ttl=254 time=31 ms --- 1.1.1.254 ping statistics --- 5 packet(s) transmitted 5 packet(s) received 0.00% packet loss round-trip min/avg/max = 31/34/46 ms PC> </pre>				

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
29	63.188000	1.1.1.253	1.1.1.254	ICMP	82	Echo (ping) request id=0x0805, seq=1/256, ttl=127 (reply in 30)
30	63.188000	1.1.1.254	1.1.1.253	ICMP	82	Echo (ping) reply id=0x0805, seq=1/256, ttl=255 (request in 29)
31	63.578000	HuaweiTe_10:39:53	HuaweiTe_a2:1f:12	PPP L...	60	Echo Request
32	63.578000	HuaweiTe_a2:1f:12	HuaweiTe_10:39:53	PPP L...	60	Echo Reply
33	64.219000	1.1.1.253	1.1.1.254	ICMP	82	Echo (ping) request id=0x0806, seq=2/512, ttl=127 (reply in 34)
34	64.219000	1.1.1.254	1.1.1.253	ICMP	82	Echo (ping) reply id=0x0806, seq=2/512, ttl=255 (request in 33)
35	65.250000	1.1.1.253	1.1.1.254	ICMP	82	Echo (ping) request id=0x0807, seq=3/768, ttl=127 (reply in 36)
36	65.266000	1.1.1.254	1.1.1.253	ICMP	82	Echo (ping) reply id=0x0807, seq=3/768, ttl=255 (request in 35)
37	66.297000	1.1.1.253	1.1.1.254	ICMP	82	Echo (ping) request id=0x0808, seq=4/1024, ttl=127 (reply in 38)
38	66.297000	1.1.1.254	1.1.1.253	ICMP	82	Echo (ping) reply id=0x0808, seq=4/1024, ttl=255 (request in 37)
39	67.219000	HuaweiTe_a2:1f:12	HuaweiTe_10:39:53	PPP L...	60	Echo Request
40	67.235000	HuaweiTe_10:39:53	HuaweiTe_a2:1f:12	PPP L...	60	Echo Reply
41	67.328000	1.1.1.253	1.1.1.254	ICMP	82	Echo (ping) request id=0x0809, seq=5/1280, ttl=127 (reply in 42)
42	67.344000	1.1.1.254	1.1.1.253	ICMP	82	Echo (ping) reply id=0x0809, seq=5/1280, ttl=255 (request in 41)
43	73.578000	HuaweiTe_10:39:53	HuaweiTe_a2:1f:12	PPP L...	60	Echo Request
44	73.578000	HuaweiTe_a2:1f:12	HuaweiTe_10:39:53	PPP L...	60	Echo Reply
45	75.625000	HuaweiTe_a2:1f:12	HuaweiTe_10:39:53	PPP L...	60	Echo Request
46	75.641000	HuaweiTe_10:39:53	HuaweiTe_a2:1f:12	PPP L...	60	Echo Reply
47	83.578000	HuaweiTe_10:39:53	HuaweiTe_a2:1f:12	PPP L...	60	Echo Request
48	83.578000	HuaweiTe_a2:1f:12	HuaweiTe_10:39:53	PPP L...	60	Echo Reply
49	84.031000	HuaweiTe_a2:1f:12	HuaweiTe_10:39:53	PPP L...	60	Echo Request
50	84.031000	HuaweiTe_10:39:53	HuaweiTe_a2:1f:12	PPP L...	60	Echo Reply
51	87.473000	HuaweiTe_a2:1f:12	HuaweiTe_10:39:53	PPP L...	60	Echo Request

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 7

Enter a keyword.		Add Filter																			
Index	Name	Descri...	Tag	VLAN ID	Sourc...	Destin...	Sourc...	Destin...	User	Service	Applic...	Sched...	Action	Conte...	Hits	Enable	Edit				
☐ 1	trust-to-untrust			any	trust	untrust	10.0.3...	any	any	any	any	any	Permit		10 Re...	☒					
2	default	This is...		any	any	any	any	any	any	any	any	any	Deny		42 Re...	☒					

```
#
nat-policy
rule name nat-policy-1
source-zone trust
egress-interface Dialer1
source-address 10.0.3.0 mask 255.255.255.0
action source-nat easy-ip
rule name GuideNat1770264999107
egress-interface Dialer1
action source-nat easy-ip
#
return
```

Modify GigabitEthernet Interface

Interface Name

GigabitEthernet1/0/2 *

Alias

Virtual System?

public *

Zone

untrust

Mode?

☒ Routing
☐ Switching
☐ Bypass
☐ Interface Pair

IPv4 IPv6

Connection Type

☐ Static IP
☐ DHCP
☒ PPPoE

User Name

user

Password

••••••••

Note: It is recommended that the password contain at least three of the following character types: uppercase letters, lowercase letters, digits, and special characters.

Disconnection Type

☒ Always online
☐ Disconnect after specified idle period

☒ Automatically obtain IP address
☐ Specify IP address:

IP Address

*<1-65535>seconds

OK Cancel

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 7

Modify DHCP Server
✕

Interface Name

Type

Service Type

IP Address Range

Subnet Mask

Default Gateway?

DNS Service

Primary DNS Server

Secondary DNS Server

☐ **Advanced**

☒ IPv4
☒ Server

☐ Use System DNS Settings

☐ IPv6
☐ Relay

-

☒ Specify

OK

Cancel

Interface List

Interface Name	Zone	IP Address	Connection ...	VLAN/VXL...	Mode	State			Enable	Edit
						Physical	IPv4	IPv6		
GE0/0/0(GE0/METH)	trust (default)	192.168.0.1	Static IP (IPv4) Static IP (IPv6)		Routing	↓	↓	↓	✓	
GE1/0/0	trust (public)	192.168.10.1	Static IP (IPv4) Static IP (IPv6)		Routing	↑	↑	↓	✓	
GE1/0/1	trust (public)	10.0.3.1	Static IP (IPv4) Static IP (IPv6)		Routing	↑	↑	↓	✓	
GE1/0/2	untrust (public)	1.1.1.253	PPPoE (IPv4) Static IP (IPv6)		Routing	↑	↑	↓	✓	
GE1/0/3	-- NONE -- (public)	---	Static IP (IPv4) Static IP (IPv6)		Routing	↓	↓	↓	✓	
GE1/0/4	-- NONE -- (public)	---	Static IP (IPv4) Static IP (IPv6)		Routing	↓	↓	↓	✓	
GE1/0/5	-- NONE -- (public)	---	Static IP (IPv4) Static IP (IPv6)		Routing	↓	↓	↓	✓	
GE1/0/6	-- NONE -- (public)	---	Static IP (IPv4) Static IP (IPv6)		Routing	↓	↓	↓	✓	
Virtual-If0	-- NONE -- (public)	---				↑	↑			

11 | Page

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 7

```
drop illegal-mac alarm
#
wlan ac-global carrier id other ac id 0
#
set cpu-usage threshold 80 restore 75
#
ip pool pool1
gateway-list 1.1.1.254
network 1.1.1.0 mask 255.255.255.0
#
aaa
authentication-scheme default
authentication-scheme system-a
authorization-scheme default
authorization-scheme system-a
accounting-scheme default
domain default
domain default_admin
domain system
authentication-scheme system-a
authorization-scheme system-a
local-user user password cipher %$%$/>k_#dFFoTTril.^-`Q3A9R-%$%$
local-user user service-type ppp
local-user admin password cipher %$%$K8m.Nt84DZ}e#<0`8bmE3Uw}%$%$
local-user admin service-type http
#
firewall zone Local
priority 15
#
interface Virtual-Templat1
ppp authentication-mode pap domain system
remote address pool pool1
ppp ipcp dns 9.9.9.9
ip address 1.1.1.254 255.255.255.0
#
```

DEPARTMENT OF TELECOMMUNICATION ENGINEERING
BACHELOR OF SCIENCE IN CYBER SECURITY
MEHRAN UNIVERSITY OF ENGINEERING & TECHNOLOGY, JAMSHORO
NETWORK SECURITY (CYS360)
(6TH SEMESTER, 3RD Year) LAB EXPERIMENT # 7

```
#
interface Virtual-Template1
  ppp authentication-mode pap domain system
  remote address pool pool1
  ppp ipcp dns 9.9.9.9
  ip address 1.1.1.254 255.255.255.0
#
interface GigabitEthernet0/0/0
#
interface GigabitEthernet0/0/1
  pppoe-server bind Virtual-Template 1
#
interface GigabitEthernet0/0/2
#
[ISP-router]
```